1. If $\Delta ABC \sim \Delta DCB$

then, $\dfrac{AB}{DC} = \dfrac{BC}{CB} = \dfrac{AC}{DB}$

Now, $\dfrac{AB}{DC} = \dfrac{AC}{DB}$

$AB \times DB = AC \times DC$

2. given $A(0,0)$ {coordinates of origin is (0,0)}
and $B(a,b)$

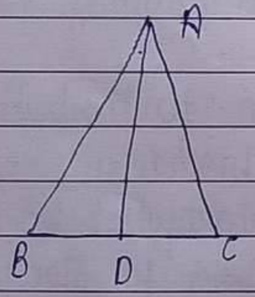
$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$AB = \sqrt{(a-0)^2 + (b-0)^2}$

$AB = \sqrt{a^2 + b^2}$ Ans

~~AB =~~ a

3.



given $\angle BAC = \angle ADC$

In ΔABC and ΔACD

$\angle BAC = \angle ADC$ (given)

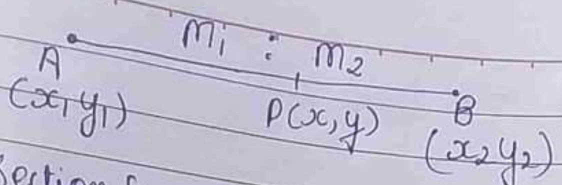
$\angle C = \angle C$ (common)

So, $\Delta ABC \sim \Delta DAC$ (By AA)

Now $\dfrac{AB}{DA} = \dfrac{BC}{AC} = \dfrac{AC}{DC}$

So, $\dfrac{CA}{CB} = \dfrac{DA}{AB} = \dfrac{DC}{AC}$ Ans

4.

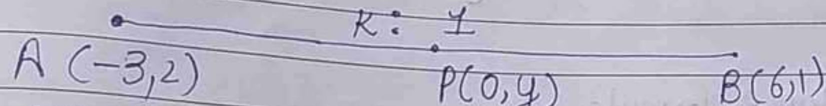


Section formula,

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

$$y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

6.



Let y-axis divide this line segment in $k:1$

By section formula, $x = \frac{k(6) + 1(-3)}{k+1}$

$$0 = \frac{6k - 3}{k+1}$$

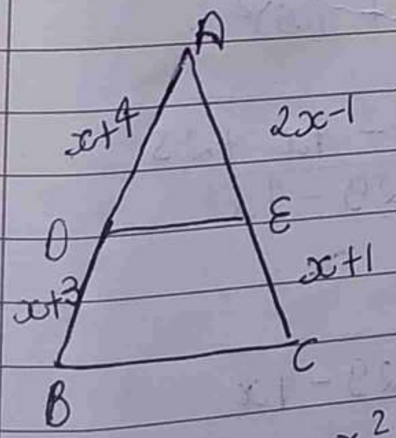
$$0 = 6k - 3$$

$$0 = 3(2k-1)$$

$$k = \frac{1}{2}$$

So, the ratio is $1:2$

7.



given $DE \parallel BC$

So, $\frac{AD}{DB} = \frac{AE}{EC}$ (By BPT)

$$\frac{x+4}{x+3} = \frac{2x-1}{x+1}$$

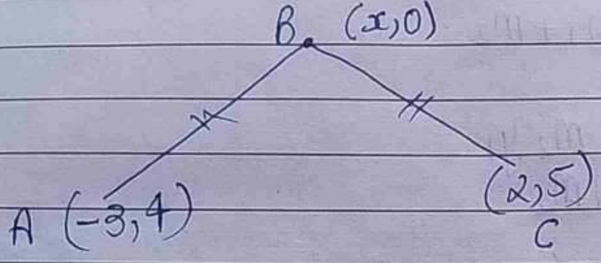
$$(x+1)(x+4) = (2x-1)(x+3)$$

$$x^2 + 4x + x + 4 = 2x^2 + 6x - x - 3$$

$$x^2 + \cancel{5x} + 4 = 2x^2 + \cancel{5x} - 3$$

$$x^2 - 2x^2 + 4 + 3 = 0$$
$$-x^2 + 7 = 0$$
$$+x^2 = +7$$
$$x = \sqrt{7}$$

8.



By distance formula

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$AB = \sqrt{(x+3)^2 + (4)^2}$$

$$AB = \sqrt{(x+3)^2 + 16}$$

$$AB = \sqrt{x^2 + 9\cancel{x} + 6x + 16}$$
$$AB = \sqrt{x^2 + 25 + 6x}$$

Now By distance formula $BC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$BC = \sqrt{(x-2)^2 + (5)^2}$$

$$BC = \sqrt{x^2 + 4 - 4x + 25}$$
$$BC = \sqrt{x^2 + 29 - 4x}$$

Now given that $AB = BC$

$$\sqrt{x^2 + 25 + 6x} = \sqrt{x^2 + 29 - 4x}$$

squaring both side

$$\cancel{x^2} + 25 + 6x = \cancel{x^2} + 29 - 4x$$
$$6x + 4x = 29 - 25$$

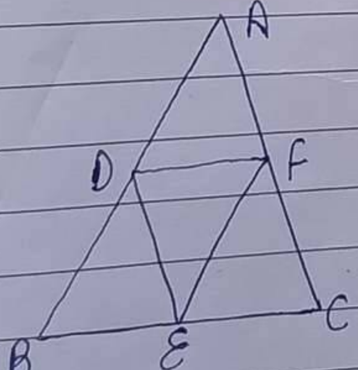
$$10x = 4$$
$$x = \frac{\cancel{4}^{\,2}}{\cancel{10}_{\,5}}$$
$$x = \frac{2}{5} \text{ units}$$

10. Let $A(2,2)$ $B(4,4)$ $C(2,6)$
x_1, y_1 x_2, y_2 x_3, y_3

Now,
$$\begin{array}{cccc} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{array}$$

$$\text{Area} = \frac{1}{2} \left| (x_1y_2 + x_2y_3 + x_3y_1) - (x_2y_1 + x_3y_2 + x_1y_3) \right|$$
$$= \frac{1}{2} \left| (8 + 24 + 4) - (8 + 8 + 12) \right|$$
$$= \frac{1}{2} \left| 36 - 28 \right|$$
$$= \frac{1}{2} \left| 8 \right|$$
$$= \frac{1}{\cancel{2}} \times \cancel{8}\,4$$
$$= 4 \text{ unit sq.}$$

5.



Given - D, E, and F are mid points of the sides AB, BC and CA

To find $\frac{ar(\Delta DEF)}{ar(\Delta ABC)}$

We know that line segment joining mid points of any two sides of a triangle is half the third side and parralel to it.

$\therefore FD = \frac{1}{2} BC, \quad ED = \frac{1}{2} AC, \quad EF = \frac{1}{2} AB$

In ΔABC and ΔEFD

$\frac{AB}{EF} = \frac{BC}{FD} = \frac{AC}{ED} = 2 \quad -(1)$

$\Delta ABC \sim \Delta EFD$ (by SSS)

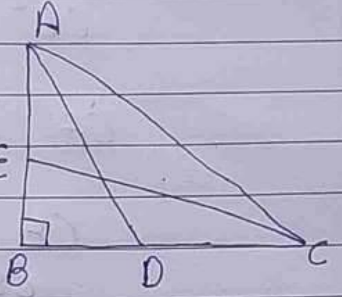
If two triangle are similar then ratio of their areas is equal to square of their corresponding sides

$\therefore \frac{ar(\Delta ABC)}{ar(\Delta EFD)} = \left(\frac{AB}{EF}\right)^2 = 4$

from (1) $\frac{ar(\Delta EFD)}{ar(ABC)} = \frac{1}{4}$

9. given :- ABC is a right triangled at B, AD and CE are two median drawn from A and C respective-ly. $AC = 5cm$ and $AD = \frac{3\sqrt{5}}{2}$ cm

To find :- length of CE



Since, ΔABD is right triangle at B, So,

$$AD^2 = AB^2 + BD^2$$

$$AD^2 = AB^2 + \left(\frac{BC}{2}\right)^2 \quad (\because BD = DC)$$

$$AD^2 = AB^2 + \frac{1}{4}BC^2 \quad -\text{(i)}$$

Again ΔBCE is right angled at B

$$\because CE^2 = BC^2 + BE^2$$

$$CE^2 = BC^2 + \left(\frac{AB}{2}\right)^2 \quad \{BE = EA\}$$

$$CE^2 = BC^2 + \left(\frac{AB}{2}\right)^2$$

$$CE^2 = BC^2 + \frac{1}{4}AB^2 \quad -\text{(ii)}$$

Adding (i) & (ii)

$$AD^2 + CE^2 = AB^2 + \frac{1}{4}BC^2 + BC^2 + \frac{1}{4}AB^2$$

$$AD^2 + CE^2 = \frac{5}{4}(AB^2 + BC^2)$$

$$AD^2 + CE^2 = \frac{5}{4}AC^2$$

$$\begin{array}{r} 25 \\ 5 \\ \hline 125 \end{array}$$

2

5

$$\left(\frac{3\sqrt{5}}{2}\right)^2 + CE^2 = \frac{5}{4} \times 25$$

$$\frac{45}{4} + CE^2 = \frac{125}{4}$$

$$CE^2 = \frac{125}{4} - \frac{45}{4}$$

$$CE^2 = \cancel{\frac{80}{4}}\,20$$

$$CE = \sqrt{20}$$

$$CE = 2\sqrt{5}$$